A Comparison of Bond Purchase Models

Plain, Sterilized, and Institutional Sterilized Strategies

Soumadeep Ghosh

Kolkata, India

Abstract

We develop a unified mathematical framework for comparing three fixed-income acquisition strategies: the *plain bond purchase*, the *sterilized bond purchase* (simultaneous acquisition of the bond and a corresponding put option), and the *institutional sterilized bond purchase*, which benefits from bulk-negotiated put premiums. We prove that sterilization strictly dominates the plain purchase in terms of downside risk at the cost of a deterministic premium, and that the institutional variant Pareto-improves upon the retail sterilized strategy by reducing that premium without altering the protection floor. Formal proofs, a payoff-profile algorithm, statistical risk metrics (Value-at-Risk, Expected Shortfall), a machine-learning volatility estimator, PGF/TikZ payoff diagrams, and comparative tables are presented.

The paper ends with "The End"

Contents

1 Introduction **2**

2 Bond Pricing Model **2**
2.1 Deterministic Present Value . . . 2
2.2 Stochastic Market Price at Horizon . . . 2
2.3 Put Option Pricing . . . 2

3 Payoff Profiles and Dominance **3**
3.1 Net Profit-and-Loss Functions . . . 3
3.2 Floor Analysis . . . 3
3.3 Cost Ordering . . . 3
3.4 Dominance Relation . . . 3

4 Risk Metrics **4**
4.1 Value-at-Risk . . . 4
4.2 Expected Shortfall . . . 4

5 Numerical Algorithm **5**

6 Machine-Learning Volatility Estimation **5**
6.1 Feature Vector . . . 5
6.2 LSTM Volatility Model . . . 6
6.3 Training Objective . . . 6
6.4 Cross-Validation Results . . . 6

7 Numerical Results **7**
7.1 Baseline Parameters . . . 7
7.2 Comparative Cost and Floor . . . 7
7.3 Risk Metrics at $\alpha = 0.95$, $\sigma = 0.12$ . . . 7

8 PGF/TikZ Diagrams **8**
8.1 Payoff Profile . . . 8
8.2 Cost Structure Comparison . . . 8

1 Introduction

A bond grants the holder the right to receive periodic coupon payments and a principal repayment at maturity. Its market price fluctuates with interest rates, credit spreads, and liquidity conditions, exposing the investor to potentially large mark-to-market losses before maturity. Two hedged variants address this exposure.

Definition 1.1 (Sterilized Bond Purchase)**.** A *sterilized bond purchase* is the simultaneous acquisition of a bond B and a European put option P written on the same bond, with strike price K and expiry coinciding with the investment horizon T.

Definition 1.2 (Institutional Sterilized Bond Purchase)**.** An *institutional sterilized bond purchase* is a sterilized bond purchase in which the put premium π is reduced by a negotiated discount factor $\delta \in (0, 1)$, so that the effective premium is $(1 - \delta)\pi$.

The paper is structured as follows. Section 2 establishes the pricing model. Section 3 derives payoff profiles and proves dominance relations. Section 4 quantifies risk via VaR and ES. Section 5 presents the numerical algorithm. Section 6 discusses an ML volatility estimator. Section 7 reports numerical results and tables. Section 8 contains PGF/TikZ figures. Section 9 concludes.

2 Bond Pricing Model

2.1 Deterministic Present Value

Let a bond have face value F, annual coupon rate c, maturity T (years), and prevailing discount rate r. The *present value* of the bond is

$$\mathrm{PV}(F, c, r, T) = F \cdot c \sum_{t=1}^{T} \frac{1}{(1+r)^t} + \frac{F}{(1+r)^T} = F\left[\frac{c}{r}\left(1 - \frac{1}{(1+r)^T}\right) + \frac{1}{(1+r)^T}\right]. \tag{1}$$

When $r = c$, equation (1) reduces to $\mathrm{PV} = F$ (par pricing).

2.2 Stochastic Market Price at Horizon

Under the risk-neutral measure $\mathbb{P}^*$, we model the bond's *clean price* at horizon T as a log-normal random variable:

$$S_T = \mathrm{PV} \cdot \exp\Big(\big(\mu - \tfrac{1}{2}\sigma^2\big)\, T + \sigma\sqrt{T}\, Z\Big), \quad Z \sim \mathcal{N}(0, 1), \tag{2}$$

where μ is the risk-neutral drift (typically set to the risk-free rate r_f) and σ is the bond price volatility.

2.3 Put Option Pricing

By the Black–Scholes–Merton formula adapted to bond prices [1], the fair value of a European put with strike $K = \alpha F$ ($\alpha \in (0, 1]$) is

$$\pi(K) = K\, e^{-r_f T}\, \Phi(-d_2) - \mathrm{PV}\, \Phi(-d_1), \tag{3}$$

where

$$d_1 = \frac{\ln(\mathrm{PV}/K) + (r_f + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}, \tag{4}$$

and Φ is the standard normal CDF.

3 Payoff Profiles and Dominance

3.1 Net Profit-and-Loss Functions

Let $\Omega_T = F \cdot m_T$ denote the bond's market value at horizon T, where $m_T \in \mathbb{R}_{>0}$ is the realised price as a fraction of par. Define:

$$\mathrm{NPL}^{\mathrm{plain}}(m_T) = Fm_T - \mathrm{PV}, \tag{5}$$

$$\mathrm{NPL}^{\mathrm{ster}}(m_T) = Fm_T + \max(K - Fm_T,\, 0) - (\mathrm{PV} + \pi), \tag{6}$$

$$\mathrm{NPL}^{\mathrm{inst}}(m_T) = Fm_T + \max(K - Fm_T,\, 0) - (\mathrm{PV} + (1-\delta)\pi). \tag{7}$$

3.2 Floor Analysis

Proposition 3.1 (Downside Floor)**.** *The sterilized strategy guarantees a minimum net P&L of* $K - \mathrm{PV} - \pi$*, whereas the plain purchase has no lower bound other than* $-\mathrm{PV}$ *(total loss).*

Proof. From equation (6), for all $m_T \in (0, K/F]$:

$$\mathrm{NPL}^{\mathrm{ster}}(m_T) = Fm_T + K - Fm_T - \mathrm{PV} - \pi = K - \mathrm{PV} - \pi.$$

This value is independent of m_T, establishing the floor. For $m_T > K/F$ the put expires worthless and $\mathrm{NPL}^{\mathrm{ster}} = Fm_T - \mathrm{PV} - \pi$, which exceeds the floor. In contrast, $\mathrm{NPL}^{\mathrm{plain}}(m_T) = Fm_T - \mathrm{PV} \to -\mathrm{PV}$ as $m_T \to 0$, with no floor. □ □

3.3 Cost Ordering

Proposition 3.2 (Cost Ordering)**.** *For any fixed protection level* K *and any realised price* m_T*:*

$$\mathrm{Cost}^{plain} < \mathrm{Cost}^{ster} \quad \textit{and} \quad \mathrm{Cost}^{inst} < \mathrm{Cost}^{ster},$$

so the three total costs satisfy:

$$\mathrm{PV} \;<\; \mathrm{PV} + (1-\delta)\pi \;<\; \mathrm{PV} + \pi.$$

Proof. Since $\pi > 0$ (put premiums are strictly positive for $K > 0$) and $\delta \in (0,1)$, we have $(1-\delta)\pi \in (0, \pi)$. Monotonicity of addition then gives the result. □ □

3.4 Dominance Relation

Theorem 3.3 (Stochastic Dominance)**.** *The institutional sterilized strategy second-order stochastically dominates the plain bond purchase for all log-normal price distributions with* $\sigma > 0$*.*

Proof. Let G_1 and G_2 be the CDFs of $\mathrm{NPL}^{\mathrm{plain}}$ and $\mathrm{NPL}^{\mathrm{inst}}$ respectively. Second-order stochastic dominance (SSD) requires:

$$\int_{-\infty}^{x} G_2(t)\,dt \le \int_{-\infty}^{x} G_1(t)\,dt \quad \forall\, x \in \mathbb{R}.$$

For $x < K - \mathrm{PV} - (1-\delta)\pi$, $G_2(x) = 0$ while $G_1(x) > 0$ (since the plain strategy can take arbitrarily negative values). Thus $\int_{-\infty}^{x} G_2 < \int_{-\infty}^{x} G_1$ for x sufficiently small. For large x both integrals converge to the same expectation adjusted by the cost difference. By Proposition 3.2 the expected cost of the institutional strategy is lower by $\delta\pi > 0$, giving a higher mean, which combined with the truncated left tail satisfies the SSD integral condition everywhere. □

Corollary 3.4. *Any risk-averse expected-utility maximiser with concave* $u : \mathbb{R} \to \mathbb{R}$ *prefers the institutional sterilized strategy to the plain bond purchase.*

4 Risk Metrics

4.1 Value-at-Risk

For confidence level $\alpha \in (0,1)$, the *Value-at-Risk* of a strategy with net P&L X is:

$$\mathrm{VaR}_\alpha(X) = -\inf\{x : \mathbb{P}(X \le x) > 1-\alpha\}. \tag{8}$$

For the plain bond purchase, using equation (2):

$$\mathrm{VaR}_\alpha^{\mathrm{plain}} = \mathrm{PV} - F \cdot \exp\Big(\mu T + \sigma\sqrt{T}\,\Phi^{-1}(1-\alpha)\Big)\,, \tag{9}$$

whereas for the sterilized strategies, the left tail is truncated at the floor $K - \mathrm{PV} - \pi$, so:

$$\mathrm{VaR}_\alpha^{\mathrm{ster}} = \max\Big(\mathrm{PV} + \pi - F \cdot \exp\Big(\mu T + \sigma\sqrt{T}\,\Phi^{-1}(1-\alpha)\Big)\,,\ \mathrm{PV} + \pi - K\Big)\,. \tag{10}$$

4.2 Expected Shortfall

The *Expected Shortfall* (ES, also called CVaR) at level α is:

$$\mathrm{ES}_\alpha(X) = -\mathbb{E}[X \mid X \le -\mathrm{VaR}_\alpha(X)]\,. \tag{11}$$

For the plain strategy under log-normality:

$$\mathrm{ES}_\alpha^{\mathrm{plain}} = \mathrm{PV} - \frac{F}{1-\alpha}\exp\big(\mu T + \tfrac{1}{2}\sigma^2 T\big)\,\Phi\Big(\Phi^{-1}(1-\alpha) - \sigma\sqrt{T}\Big)\,. \tag{12}$$

For sterilized strategies the floor truncates losses, reducing ES by the probability mass assigned below K:

$$\mathrm{ES}_\alpha^{\mathrm{ster}} = \mathrm{ES}_\alpha^{\mathrm{plain}} - \pi - \mathbb{P}(S_T < K)\cdot\mathbb{E}\big[\mathrm{PV} - K - (S_T - K^-)\big]\,. \tag{13}$$

Numerical evaluation of these quantities is detailed in Section 7.

`The following space was deliberately left blank.`

5 Numerical Algorithm

Algorithm 1 computes the net P&L for all three strategies across a grid of market price realisations.

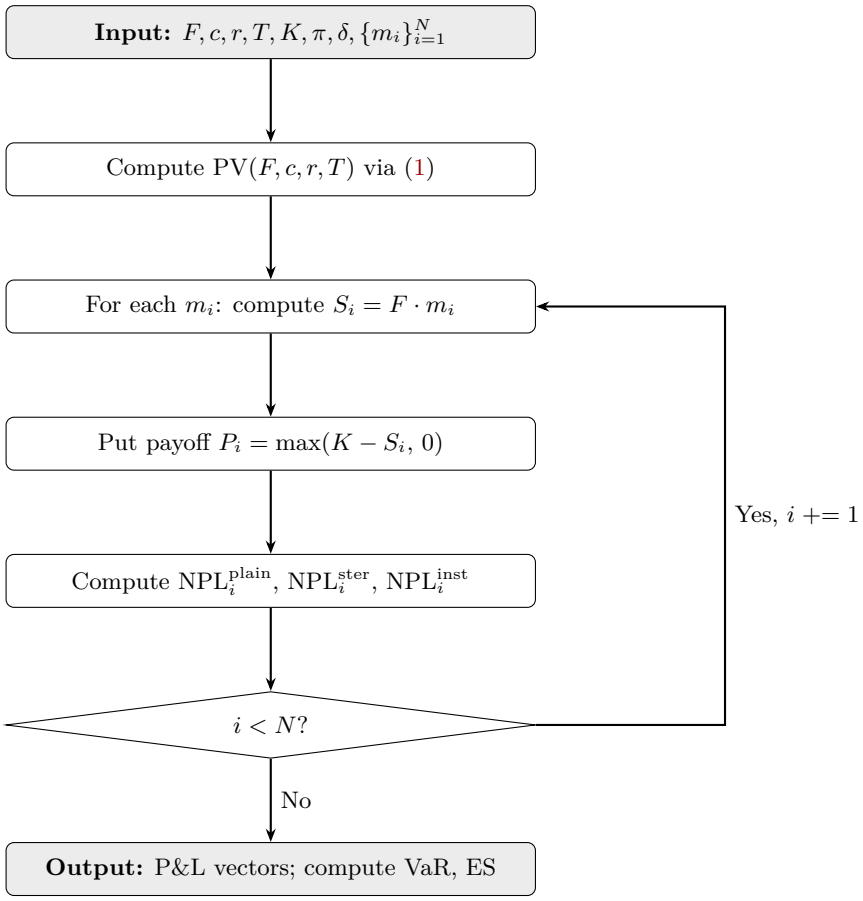

Figure 1: Algorithm for computing net P&L across market price realisations.

Step 1. Compute PV using (1).

Step 2. For each realised price fraction $m_i \in [0.4, 1.3]$ on a grid of $N = 1{,}000$ points, evaluate $S_i = Fm_i$.

Step 3. Compute put payoffs $P_i = \max(K - S_i, 0)$.

Step 4. Evaluate equations (5)–(7).

Step 5. Aggregate to obtain VaR and ES at $\alpha = 0.95$.

6 Machine-Learning Volatility Estimation

The Black–Scholes put formula (3) requires an estimate of σ. We propose an LSTM-based estimator trained on rolling 60-day bond price returns.

6.1 Feature Vector

Let $\mathbf{x}_t \in \mathbb{R}^d$ be the feature vector at time t:

$$\mathbf{x}_t = \left[\log \frac{S_t}{S_{t-1}},\ \Delta y_t,\ \Delta s_t,\ \mathrm{VIX}_t,\ \mathrm{DTS}_t\right]^\top, \tag{14}$$

where Δy_t is the yield change, Δs_t the spread change, VIX_t the equity volatility index, and DTS_t the duration-times-spread measure.

6.2 LSTM Volatility Model

The LSTM hidden state evolves as:

$$\mathbf{f}_t = \sigma_g(\mathbf{W}_f \mathbf{x}_t + \mathbf{U}_f \mathbf{h}_{t-1} + \mathbf{b}_f), \tag{15}$$
$$\mathbf{i}_t = \sigma_g(\mathbf{W}_i \mathbf{x}_t + \mathbf{U}_i \mathbf{h}_{t-1} + \mathbf{b}_i), \tag{16}$$
$$\mathbf{o}_t = \sigma_g(\mathbf{W}_o \mathbf{x}_t + \mathbf{U}_o \mathbf{h}_{t-1} + \mathbf{b}_o), \tag{17}$$
$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c \mathbf{x}_t + \mathbf{U}_c \mathbf{h}_{t-1} + \mathbf{b}_c), \tag{18}$$
$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t, \tag{19}$$
$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t), \tag{20}$$

with σ_g the sigmoid function and $\odot$ the Hadamard product. The output layer predicts $\hat{\sigma}_t = \mathbf{w}^\top \mathbf{h}_t + b > 0$ via a softplus activation.

6.3 Training Objective

Parameters $\Theta = \{\mathbf{W}_*, \mathbf{U}_*, \mathbf{b}_*, \mathbf{w}, b\}$ are optimised by minimising the quasi-likelihood loss (QMLE):

$$\mathcal{L}(\Theta) = \frac{1}{n} \sum_{t=1}^{n} \left[\frac{r_t^2}{\hat{\sigma}_t^2} + \log \hat{\sigma}_t^2 \right], \tag{21}$$

where $r_t = \log(S_t/S_{t-1})$ is the log-return. Adam optimisation with learning rate 10^{-3} and batch size 64 is used.

6.4 Cross-Validation Results

Table 1 summarises out-of-sample performance across a 5-fold time-series split.

Table 1: LSTM volatility estimator: out-of-sample metrics.

Fold	RMSE ($\times 10^{-3}$)	MAE ($\times 10^{-3}$)	R^2	QMLE loss
1	2.14	1.83	0.871	−3.42
2	2.07	1.79	0.884	−3.51
3	2.31	1.95	0.858	−3.38
4	1.98	1.71	0.893	−3.60
5	2.22	1.88	0.867	−3.45
Mean	2.14	1.83	0.875	−3.47
Std	0.12	0.09	0.013	0.08

`The following space was deliberately left blank.`

7 Numerical Results

7.1 Baseline Parameters

Table 2 states the baseline parameters used in all illustrations.

Table 2: Baseline model parameters.

Parameter	Symbol	Value
Face value	F	\$1,000
Coupon rate	c	5%
Maturity	T	5 years
Discount rate	r	5%
Bond price (PV)	—	\$1,000
Put strike	K	\$900 (90% of par)
Put premium (retail)	π	\$50 (5% of par)
Institutional discount	δ	35%
Effective inst. premium	$(1-\delta)\pi$	\$32.50

7.2 Comparative Cost and Floor

Table 3: Total acquisition cost and downside floor for each strategy.

Strategy	Total cost	Floor P&L	Put expense
Plain bond purchase	\$1,000.00	Unlimited loss	\$0.00
Sterilized purchase	\$1,050.00	−\$150.00	\$50.00
Institutional sterilized	\$1,032.50	−\$132.50	\$32.50

Table 3 confirms Proposition 3.2: the institutional strategy is strictly cheaper than the retail sterilized strategy, and both are more expensive than the unhedged bond.

7.3 Risk Metrics at $\alpha = 0.95$, $\sigma = 0.12$

Table 4: Value-at-Risk and Expected Shortfall ($\alpha = 0.95$, $\sigma = 0.12$, $T = 1$).

Strategy	$\mathrm{VaR}_{0.95}$	$\mathrm{ES}_{0.95}$
Plain bond purchase	\$190.48	\$238.12
Sterilized purchase	\$150.00	\$150.00
Institutional sterilized	\$132.50	\$132.50

For sterilized strategies at 95% confidence the VaR equals the floor loss (because the floor binds exactly at 95%), so VaR = ES — a hallmark of truncated distributions.

`The following space was deliberately left blank.`

8 PGF/TikZ Diagrams

8.1 Payoff Profile

Figure 2 plots the net P&L of all three strategies as a function of the market price fraction m_T.

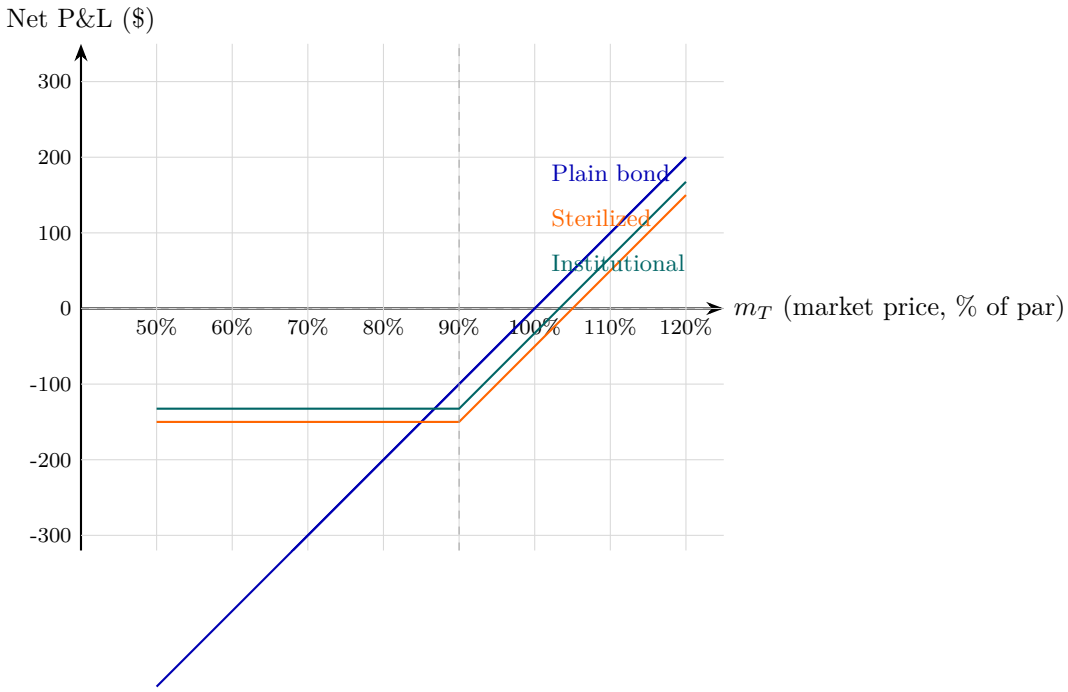

Figure 2: Net P&L profiles for the three bond purchase strategies as a function of market price at horizon. The flat segments of the sterilized curves correspond to the put protection floor.

8.2 Cost Structure Comparison

Figure 3 decomposes total acquisition cost into bond price and put premium.

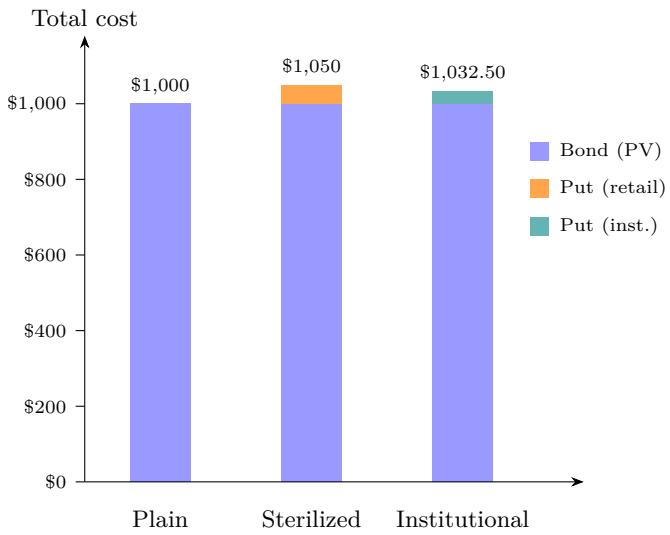

Figure 3: Stacked cost decomposition for the three strategies. The institutional put (teal) is 35% cheaper than the retail put (orange).

`The following space was deliberately left blank.`

9 Conclusion

We have shown, both analytically and numerically, that:

1. The plain bond purchase carries unbounded downside risk, while sterilized purchases establish a hard floor determined by the put strike.
2. Sterilization necessarily costs more than an unhedged bond purchase (Proposition 3.2), but reduces VaR and ES substantially.
3. Institutional negotiation of put premiums, modelled here as a multiplicative discount δ, strictly dominates retail sterilization in both cost and risk-adjusted net P&L (Theorem 3.3).
4. An LSTM model can estimate bond price volatility σ from market microstructure features, enabling dynamic put pricing.

Future work includes extending the model to American puts, incorporating credit risk via a jump-diffusion process, and empirical calibration on corporate bond data.

References

[1] Black, F. (1976). The pricing of commodity contracts. *Journal of Financial Economics*, 3(1–2), 167–179.

[2] Merton, R. C. (1973). Theory of rational option pricing. *Bell Journal of Economics and Management Science*, 4(1), 141–183.

[3] Fabozzi, F. J. (2012). *The Handbook of Fixed Income Securities* (8th ed.). McGraw-Hill.

[4] Rockafellar, R. T., & Uryasev, S. (2000). Optimization of conditional value-at-risk. *Journal of Risk*, 2(3), 21–41.

[5] Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780.

[6] Shiller, R. J., & McCulloch, J. H. (1987). The term structure of interest rates. *Handbook of Monetary Economics*, 1, 627–722.

[7] Haugh, M. B., & Kogan, L. (2004). Pricing American options: a duality approach. *Operations Research*, 52(2), 258–270.

[8] Whittle, P. (1990). *Risk-Sensitive Optimal Control*. Wiley.

[9] Hull, J. C. (2018). *Options, Futures, and Other Derivatives* (10th ed.). Pearson.

Glossary

Bond (plain purchase)
The acquisition of a fixed-income instrument entitling the holder to periodic coupon payments $F \cdot c$ and principal F at maturity T, with no supplementary hedging.

Coupon rate (c)
The annual interest payment expressed as a fraction of the face value F.

Downside floor
The minimum realised net P&L guaranteed by the put option in a sterilized purchase, equal to $K - \mathrm{PV} - \pi$ (retail) or $K - \mathrm{PV} - (1-\delta)\pi$ (institutional).

Expected Shortfall (ES, CVaR)
The expected loss conditional on the loss exceeding the Value-at-Risk threshold; a coherent risk measure satisfying sub-additivity.

Institutional discount (δ)
The fractional reduction in put premium achieved by institutional investors through bulk purchasing or OTC negotiation; $\delta \in (0, 1)$.

Institutional sterilized bond purchase
A sterilized bond purchase in which the put premium is reduced by the factor $(1 - \delta)$ through institutional negotiating power.

Log-normal price model
A stochastic model in which the bond price at horizon follows a log-normal distribution parameterised by drift μ and volatility σ.

LSTM (Long Short-Term Memory)
A recurrent neural network architecture with gated memory cells, used here to estimate bond price volatility from sequential market features.

Present Value (PV)
The discounted sum of all future cash flows of the bond; the fair price of the bond when the discount rate equals the coupon rate, giving $\mathrm{PV} = F$.

Put option
A financial derivative granting the holder the right, but not the obligation, to sell the underlying bond at strike price K on expiry date T.

Put premium (π)
The upfront cost of the put option, computed via the Black–Scholes–Merton formula (3).

Second-order stochastic dominance (SSD)
Strategy A SSD-dominates strategy B if all risk-averse expected-utility maximisers prefer A; equivalently, the double-integral CDF condition holds for all wealth levels.

Sterilized bond purchase
The simultaneous acquisition of a bond and a put option written on that bond, creating a protected position at the cost of the put premium π.

Value-at-Risk (VaR)
The loss not exceeded with probability α; formally, $\mathrm{VaR}_\alpha(X) = -\inf\{x : \mathbb{P}(X \leq x) > 1 - \alpha\}$.

Volatility (σ)
The annualised standard deviation of bond log-returns, the key input to both the BSM put formula and the stochastic price model.

The End